

The evolving landscape of drug targets

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Abstract

Since the turn of the century, drug discovery has been transformed by advances in genetics, genomics and proteomics, as well as automation, data science and the diversification of therapeutic modalities beyond synthetic small molecules and natural product derivatives. This transformation has led to the expansion of the drug target landscape, enabling advances in the treatment of many diseases, including some that previously lacked pharmacotherapies. In this Review, we map and quantify key elements of these changes over the past 25 years, such as trends in the number and type of targets through which drugs mediate their therapeutic effects, which now include 686 biomolecules modulated by 1,702 drugs. We also discuss trends we believe will continue or change, as target druggability continues to evolve with the emergence of new therapeutic modalities.

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Introduction

The molecular targets through which a drug mediates its approved therapeutic activities are commonly referred to as efficacy or mechanism of action (MoA) targets, or simply drug targets^{1,2}. Understanding the characteristics of these targets has a key role in the discovery of 'first-in-class' drugs and in clarifying opportunities to address the limitations of existing therapies, as well as in benchmarking therapeutic progress. Consequently, it is valuable to define drug targets and track the dynamics of the target landscape as drugs with new targets are approved each year.

In 2006, a landmark study identified 324 molecular drug targets, including 58 of non-human origin¹. A subsequent analysis published in 2017 reported a 2-fold increase in efficacy targets with 667 human and 189 non-human biomolecules². This increase was, in part, due to differences in the definitions of drug targets used in the two studies. In this Review, we have used a revised approach for target definition that avoids disproportionately inflating target counts related to protein complexes such as voltage-gated potassium channels, as described below and in Box 1. Complexities in the definition of drug targets are discussed in depth by Santos et al.².

In recent years, drug approvals have reached record annual levels³. Here, we examine recent trends in MoA targets, drug types (as defined in Box 1) and therapeutic areas for drugs approved by the US Food and Drug Administration (FDA), European Medicines Agency (EMA) and Japan's Pharmaceuticals and Medical Devices Agency (PMDA), building on insights from the Target Watch series of articles published between 2018 and 2025 (refs. 4–9). Notably, the completion of the first working draft from the Human Genome Project in 2000 marked a watershed moment for drug discovery, enabling systematic target identification and validation, so we place these developments in the broader context of the past 25 years.

The current landscape of drug targets

Comparison with previous analyses

Compared with the assessment of drug targets reported in 2016, we restricted the present analysis to information extracted from DrugCentral^{10,11} (sharing target information with Pharos¹²) – both resources developed within the Illuminating the Druggable Genome (IDG) project initiated by the National Institutes of Health (NIH) in 2014 (ref. 13) – to ensure a stringent and homogeneous framework for assigning drug targets (Box 1). Targets were curated from drug approval reviews and, when these were unavailable (for example, for the PMDA), from convergent evidence in the scientific literature. In this analysis, we also removed proteins that do not directly interact with a drug to mediate its therapeutic activities. For instance, instead of the 25 MoA kinases that were previously annotated for vandetanib², only 3 MoA targets are now annotated for this kinase inhibitor¹⁰.

Second, large protein complexes (≥ 5 subunits) were collapsed to a single representative protein to avoid disproportionately inflating target counts. For example, two complexes alone – mitochondrial complex I (NADH dehydrogenase, 50 subunits) and voltage-gated potassium channels (40 subunits) – account for 15% of all human efficacy targets. To address this counting bias, seven protein complexes consisting of more than five subunits were treated as single targets (Box 1). These seven complexes comprise 122 subunits total, as mapped in the UniProtKB database¹⁴ (Supplementary Table 1). For non-human targets, bacterial ATP synthase (eight subunits) was similarly treated as a single target (Supplementary Table 2).

Third, pathogenic targets were defined as unique entities regardless of species variations, reducing the count to 76 distinct pathogenic drug targets and providing a more biologically meaningful framework for classifying MoA targets. To facilitate comparisons, applying this revised overall framework to the 2017 dataset [2] yields 544 drug targets, of which 483 are human and 61 are pathogenic.

Overview of the current target landscape

The current drug target landscape (encompassing drugs approved through June 2025) comprises 686 biomolecules modulated by 1,702 drugs (Supplementary Table 3): 79% small molecules, 19% biologics and 2% other modalities (Box 1). Human biomolecules account for 90% ($n = 610$) of all drug targets (Supplementary Table 1), whereas the remaining 10% ($n = 76$) are distributed among viral, bacterial, fungal and parasitic pathogens (Supplementary Table 2).

Among human targets (Fig. 1a), enzymes are the most abundant class ($n = 209$). Four main groups – enzymes, G-protein-coupled receptors (GPCRs), ion channels and catalytic receptors (Box 1) – make up 78% ($n = 475$) of all human drug targets. Smaller target groups include transporters ($n = 23$, 4%) and nuclear hormone receptors ($n = 19$, 3%).

Among pathogen targets, two-thirds are bacterial ($n = 32$) and viral ($n = 27$) proteins. Notably, some drug targets are conserved across species, enabling broad-spectrum antimicrobial activity. For example, dihydrofolate reductase (DHFR) is targeted by trimethoprim (antibacterial), pyrimethamine (antiparasitic) and trimetrexate (antifungal), demonstrating how a single enzymatic target can be therapeutically exploited across multiple pathogen types.

Some target protein groups have been preferentially targeted by particular drug types. The most popular protein groups targeted by small-molecule drugs are enzymes ($n = 191$, representing 91% of the enzyme MoA targets), followed by ion channels ($n = 90$; 98% of the ion channel MoA targets) and the 'other proteins' category ($n = 82$, of which 67 are pathogen targets). Although GPCRs rank fourth by target count ($n = 74$), they are modulated by the largest number of small-molecule drugs ($n = 358$), reflecting extensive chemical optimization within drugs for this historically successful target class. Among the smaller protein groups, nuclear receptors (all 19 nuclear receptors) and transporters (21 of the 23 protein members) are almost exclusively targeted by small-molecule drugs.

Biologics display a markedly different target profile. Catalytic receptors ($n = 49$) constitute the largest group, followed by the diverse 'other targets' category ($n = 47$), which includes lymphocytic antigens (HLAs, CDs, BCRs and TCRs) and pathogen proteins (predominantly viral), as classified by the International Union of Basic and Clinical Pharmacology (IUPHAR) system¹⁵. Notably, biologics have emerged as highly effective modulators of endogenous peptides. Currently, 64 approved biologics target 40 of the 43 endogenous peptide targets, including growth factors, interleukins and other cytokines (Box 1), demonstrating the exceptional utility of antibody-based therapeutics for intercepting soluble signalling mediators. Other drug types (Box 1), including radioligands and inorganic salts, collectively target 36 proteins, with more than half being enzymes and GPCRs (Fig. 1b).

Having established the current landscape of drug targets, we next analyse trends in target innovation, drug approvals and drug types over the past 25 years.

Trend analysis for the twenty-first century

The release of the first working draft of the human genome in 2000 was a pivotal advance for modern drug discovery, enabling greater

Box 1 | Data collection, standardization, classification and analysis

Data sources

Drug information was primarily extracted from the DrugCentral database^{10,30} and the Target Watch series (2018–2025)^{4–9}, supplemented by manual curation from scientific literature and drug label inserts. Data collected include drug approvals by three major regulatory agencies (US Food and Drug Administration (FDA), European Medicines Agency (EMA), Pharmaceuticals and Medical Devices Agency (PMDA), drug modalities, mechanism of action (MoA) protein targets, in vitro potency values and World Health Organization (WHO) Anatomical Therapeutic Chemical (ATC) classification codes.

Drug types

Drug modalities were categorized into three types for this analysis for simplicity: small molecules; biologics (antibodies, oligonucleotides, other proteins, enzymes, fusion proteins and peptides, excluding vaccines and gene or cell therapies, which are treated separately in Box 4); and other drugs, including inorganic compounds, radioligands, complexes of two or more structures (such as dimenhydrinate and spinosad), polymers such as crofelemer (a proanthocyanidin oligomer) or heparins (a polysaccharide). We note that many of these drug categories are often treated separately, and have annotated them as such in Supplementary information.

Standardization of MoA protein targets

MoA protein targets were mapped to UniProtKB entries¹⁵ and carefully curated. Depending on the target assignment methodologies and drug approval agencies included, the count of drug targets varies significantly in other studies. For example, Lorente et al.³¹ used broad disease association scores rather than strict primary mechanism criteria, increasing G-protein-coupled receptor (GPCR) drug targets counts compared with the data described in this Review. A comprehensive discussion about the complexity of assigning drug targets is provided by Santos et al.²

Some targets represent protein complexes composed of multiple subunits; these were itemized to individual proteins for accuracy. Among all drug targets, 95% have fewer than five components, mapped as UniProt IDs. To avoid analytical bias, seven human drug targets defined as protein complexes consisting of more than five subunits were treated as single targets: mitochondrial complex I (50 subunits), voltage-gated potassium channels (40 subunits), three GABA receptor complexes (7, 13 and 16 subunits), tubulin (10 subunits) and glutamate receptors (7 subunits) (Supplementary Table 1). Likewise, one non-human target (bacterial ATP synthase) covers eight subunits as described in Supplementary Table 2.

Non-protein MoA targets

Several approved drugs bind non-protein or non-canonical protein targets, such as pre-mRNA, mRNA or peptides. Here, we annotate as MoA targets the mature proteins associated with these upstream or downstream pathways. For example, inotersen binds the *TTR* mRNA molecule to block translation, and therefore the transthyretin (TTR) protein is annotated as its MoA target. Calcitonin gene-related peptide (CGRP), a 37-amino acid neuropeptide, is bound directly

by the mAb fremanezumab, neutralizing it to prevent downstream receptor activation.

Non-human MoA protein targets

Organism-specific UniProt accession numbers were consolidated to remove redundancy due to strain variants. For example, avibactam, which targets β -lactamase, was represented without enumerating more than 1,300 strain-specific entries. Non-human targets were thus generalized without strain-specific detail.

Drug target classification

Drug targets were classified according to the [IUPHAR/BPS Guide to Pharmacology](#) system¹⁶. Protein classification guided by NC-IUPHAR (the International Union of Basic and Clinical Pharmacology Committee on Receptor Nomenclature and Drug Classification) lists seven groups of protein receptors for drugs: enzymes (protein catalysts facilitating the conversion of substrates into products)³², GPCRs³³, catalytic receptors (cell-surface proteins composed of a ligand-binding domain on the extracellular surface and a functional domain on the intracellular face of the plasma membrane, with catalytic activity separated by a single transmembrane-spanning domain of 20–25 hydrophobic amino acids)³⁴, ion channels (including ligand-gated, voltage-gated and other ion channels, which are pore-forming proteins that allow the flow of ions across cellular membranes)³⁵, nuclear hormone receptors (specialized transcription factors which bind to specific consensus sequences of DNA in the promoter region of particular target genes)³⁶, transporters (which carry solutes across cell membranes)³⁷ and other proteins³⁸. On top, we added an eighth group consisting of what the NC-IUPHAR classification system catalogues as ligands, more specifically ‘Endogenous peptide in human, mouse or rat’, such as cytokines, growth factors, immunoglobulin chains and complement factors. Many of the members of this group are proteins that bind to specific receptors, mostly catalytic receptors. For example, monoclonal antibodies such as bevacizumab target the vascular endothelial growth factor A (VEGF-A), blocking its binding to VEGF receptor tyrosine kinases. In the NC-IUPHAR classification system, kinases represent a major drug family², spanning the catalytic receptor and enzyme groups for the most part^{32,34}.

Therapeutic areas

Drugs were mapped to the WHO ATC Classification System codes, enabling hierarchical categorization by organ/system (level 1), therapeutic indication (level 2), pharmacological action (level 3), chemical class (level 4) and chemical substance (level 5).

The 14 ATC level 1 categories are: A, alimentary tract and metabolism; B, blood and blood forming organs; C, cardiovascular system; D, dermatologicals; G, genito-urinary system and sex hormones; H, systemic hormonal preparations, excluding sex hormones and insulins; J, anti-infectives for systemic use; L, antineoplastic and immunomodulating agents; M, musculo-skeletal system; N, nervous system; P, antiparasitic products, insecticides and repellents; R, respiratory system; S, sensory organs; and V, various. Whereas many recent pharmacological actions specify molecular targets, earlier codes often reflect phenotypic classifications.

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CAS registration dates

CAS Registry Numbers for 997 drugs approved between 2000 and 2025 (June) were retrieved from DrugCentral and literature sources; that is, 90% of the small-molecule drugs, 97% of the biologics and 66% of the other drugs approved in this time frame. Using the CAS numbering system, approximate registration dates were interpolated,

enabling analysis of time intervals between compound registration and drug approval.

Analysis methods and tools

All figures were generated using R version 4.3.3 package 'ggplot2' (see Related links).

understanding of both the extent to which human genes are linked to disease and the extent to which they had already been and could be targeted by drugs. Here, we use this timepoint as the basis for comparing the modern era of drug target innovation (from 2000 to 2025) with the preceding decades, and highlight key trends in drug discovery for small molecules, biologics and other drug types (Box 1). We acknowledge that the time needed to progress from the identification of novel drug targets to the approval of the first drug that modulates it may be a decade or more, meaning that the impact of the availability of the draft human genome data in 2000 on the approval of drugs with novel targets would not be immediately apparent. However, major changes in drug discovery technologies and their application in both target identification and the identification of novel drug candidates based, in part, on genetic data began before 2000 and were already leading to first-in-class drugs around the turn of the century. For example, the pioneering molecularly targeted cancer drug imatinib, which changed chronic myeloid leukaemia from a rapidly fatal disease to a manageable condition, was approved in 2001. We thus consider this timepoint to be a suitable approximate choice around which to make comparisons about trends in the target landscape and approaches to drug discovery in the two eras.

This temporal analysis encompasses three complementary perspectives: comparative analysis of targets before and after 2000, an examination of trends in drug potency, and a year-by-year tracking of approval and target introductions.

Major shifts in therapeutic targets and drug discovery

A comparison between the targets of drugs approved before and after 2000 reveals important shifts in the target landscape. Since 2000, drugs that introduced a further 312 MoA targets have been approved, representing an 84% increase from the 375 MoA targets for drugs approved before 2000. For simplicity, this discussion focuses exclusively on small molecules and biologics, which together account for 97% ($n = 1,649$) of all drugs and 98% ($n = 671$) of all targets.

Among pre-2000 targets (Fig. 2a), enzymes and ion channels are the largest groups, followed by GPCRs and pathogenic targets (49 of 60 proteins classified as 'other proteins'). The largest groups among the post-2000 targets are enzymes, catalytic receptors and other proteins (non-traditional and pathogenic protein targets), collectively comprising 70% of novel targets in this period (Fig. 2a).

One of the trends contributing to enzymes maintaining their position as the most targeted group of proteins post 2000 is the emergence of protein kinases as highly successful drug targets in the past 25 years, especially in oncology. Building on the breakthrough approval of the BCR-ABL kinase inhibitor imatinib in 2001, many other kinases linked to cancer have been targeted, such as RAF, MEK and CDK4/6, often with multiple generations of inhibitors to combat treatment resistance¹⁶. Kinases such as those in the JAK family have also become successful

targets for immune disorders¹⁷. As of 2025, there are 136 approved kinase inhibitors (78% are small molecules) targeting 67 distinct kinases (Supplementary Table 4). Within the IUPHAR classification system, 37 human kinase targets are categorized as enzymes and 28 as catalytic receptors, whereas two viral kinases are classified as 'other proteins'.

Catalytic receptor kinases, located at the cell surface, can be targeted by both small molecules (which typically access intracellular catalytic domains) and biologics (which bind extracellular regions), and have become popular targets for cancer therapies. Among this group, 12 cell-surface kinases are targeted by biologics ($n = 28$ drugs, predominantly monoclonal antibodies (mAbs) and peptides), whereas 25 have been targeted by small molecules ($n = 53$), with some targeted by both modalities. A prominent example is the epidermal growth factor receptor (EGFR; also known as human epidermal growth factor receptor 1 (HER1)). This is a MoA target for ten small-molecule drugs – ranging from the pioneering EGFR kinase inhibitor gefitinib approved in 2002 to the latest member of the class, lazertinib, approved in 2024 – and for four mAbs (cetuximab, panitumumab, necitumumab and amivantamab). Another leading example is HER2, which was first targeted by the mAb trastuzumab in 1998, and has since been targeted by six small-molecule kinase inhibitors, four mAbs and two antibody–drug conjugates (ADCs).

The most striking shifts include a 7.2-fold increase in the number of endogenous peptide targets and a 2-fold increase in catalytic receptor targets over the past 25 years. Conversely, ion channels and nuclear hormone receptors have shown minimal expansion as novel therapeutic targets: the total number of nuclear hormone receptor targets increased from 18 to 19 and the number of ion channel targets increased from 83 to 91. Notably, GPCRs – historically, the most successful drug target class based on the number of drugs targeting them (~29% of drugs pre 2000) – rank fourth in terms of the proportion of novel targets introduced post 2000 (~13%) and third in terms of the proportion of drugs targeting GPCRs overall introduced post 2000 (~21%).

Powered by advances with mAb technologies in particular, biologics have reshaped the landscape of drug targets in the past 25 years. Whereas pre-2000 targets were predominantly modulated by small molecules, with a 6:1 ratio to biologics (330 targets modulated by small molecules and 54 targeted by biologics), the ratio for the targets introduced post 2000 shifted dramatically to 1.35:1 (188 targets modulated by small molecules and 140 targets modulated by biologics), reflecting a much more similar distribution of targets between the two drug types. The number of drug targets shared by both small molecules and biologics has almost doubled since 2000, increasing from 14 to 24.

Notably, biologics have enabled the pursuit of extracellular and cell-surface proteins, particularly endogenous peptides, as drug targets. Among the biologics approved since 2000, 57 modulate endogenous peptides – compared with only one small molecule, voxelotor,

which binds haemoglobin subunit- α for the treatment of sickle cell disease¹⁸ (Fig. 2a).

Although biologics have been explored for ion channels, only one such drug has so far been approved: ziconotide, a peptide inhibitor of the voltage-dependent N-type calcium channel subunit $\alpha 1B$ (*CACNA1B*), which was approved in 2004 for pain management¹⁹. Due to cellular access limitations, biologics such as mAbs remain suboptimal platforms for targeting transporters (with functional binding sites residing in intracellular domains) and nuclear hormone receptors. One biologic targets transporters: bulevirtide, a peptide acting on the sodium/bile acid cotransporter (*SLC10A1*), approved in 2020 to treat hepatitis D virus infection²⁰. No biologics were approved for nuclear hormone receptors, consistent with the total cytosolic/nuclear subcellular location for members of this family, which are therefore outside the scope of interactions with classical mAbs.

In vitro potency of small-molecule drugs

Beyond the shifts in target distribution, examining the potency of small-molecule drugs provides insight into the quality and precision of modern drug discovery efforts. We analysed in vitro potency data from 1,625 drug–target interactions, encompassing 985 small-molecule drugs (approved through June 2025) and 415 MoA targets.

The overall mean potency is 7.7 (s.d. = 1.6) (expressed as $-\log_{10}[M]$), corresponding to 50 nM. Drugs approved after 2000 demonstrate notably higher potency compared with pre-2000 drugs: mean potency of 8.2 (s.d. = 1.3) versus 7.4 (s.d. = 1.7; $-\log_{10}[M]$), respectively (Fig. 2b). Kinase inhibitors are particularly noteworthy, exhibiting mean potencies of 8.4 (s.d. = 1.0), which substantially contribute to the enhanced potencies observed across enzyme and catalytic receptor targets. Indeed, enzymes, catalytic receptors, GPCRs, transporters and the ‘other proteins’ category are all targeted by significantly

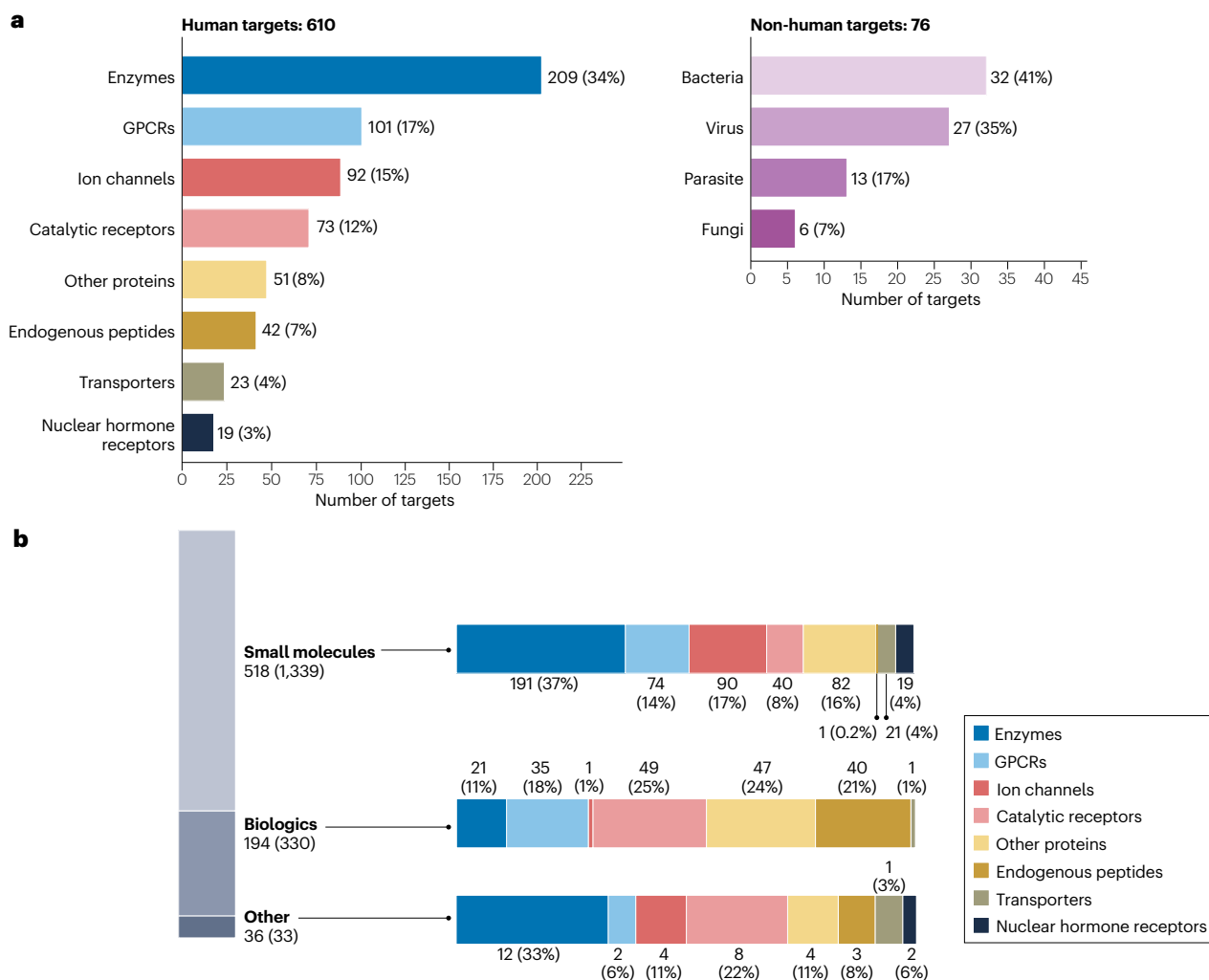


Fig. 1 | The current drug target landscape. a, Number and distribution of protein drug targets of human and non-human origins based on the protein classification system of the International Union of Basic and Clinical Pharmacology (IUPHAR) (Box 1). **b**, The proportions of protein targets (human and non-human) in each protein group according to the drug types: small-molecule drugs, biologics and other drugs. Small-molecule drugs target distinct protein classes compared with biologics. For example, enzymes represent 37%

of the small-molecule drug targets and 11% of the targets for biologics, whereas catalytic receptors represent 25% of the targets for biologics and 8% of the targets for small-molecule drugs. A small proportion of targets are modulated by more than one drug type. For example, of the 191 enzyme targets for small-molecule drugs, 177 are only modulated by small-molecule drugs, 10 are also modulated by biologics, 3 are also modulated by other drugs and 1 is modulated by all three drug types. GPCR, G-protein-coupled receptor.

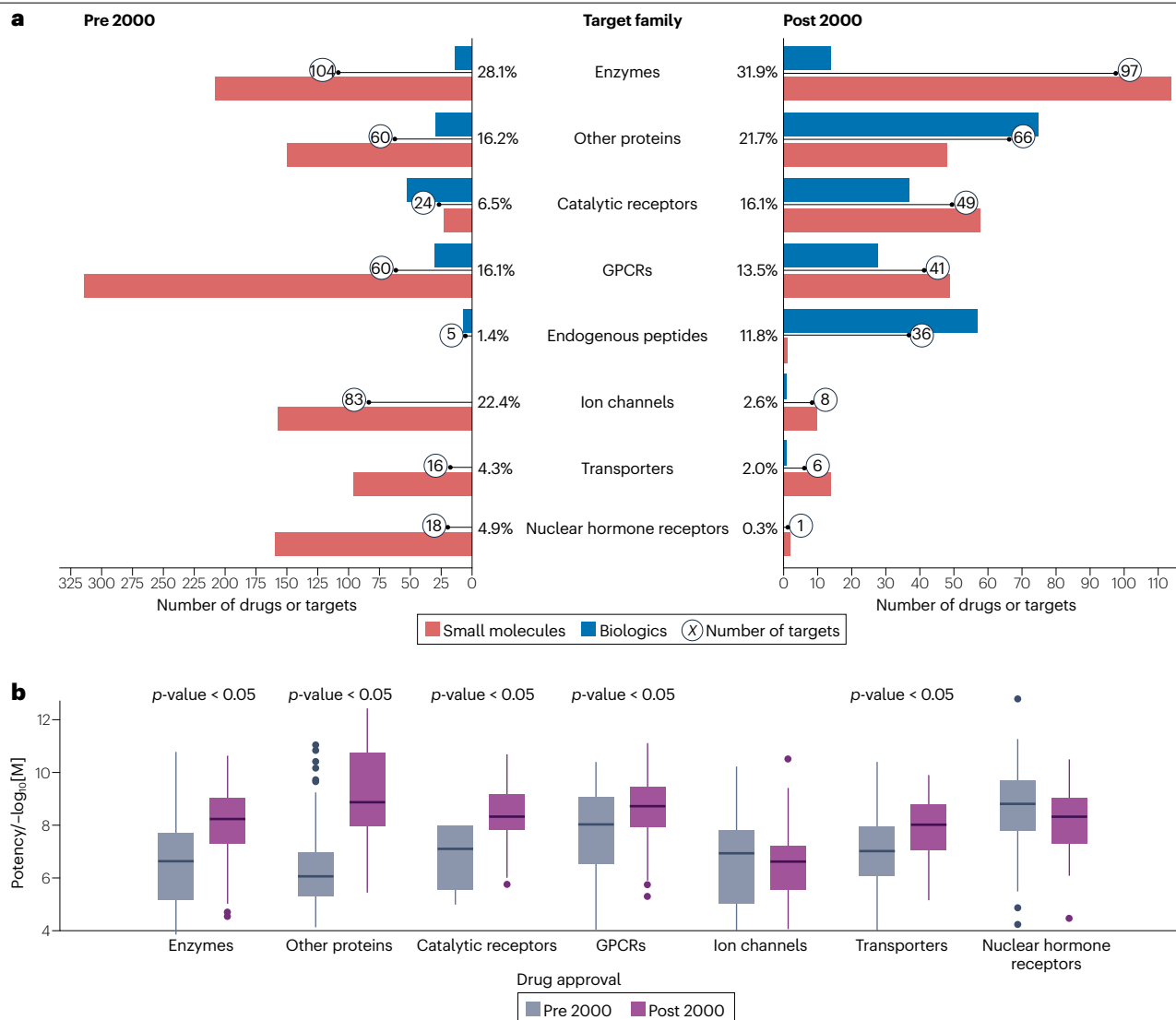


Fig. 2 | Distribution of drug types, targets and potency across protein groups and periods. **a**, Numbers of approved drugs (bars) and mechanism of action (MoA) protein targets (black lines) categorized by International Union of Basic and Clinical Pharmacology (IUPHAR) protein groups for targets introduced before 2000 (modulated by MoA-annotated drugs current as of June 2025) and after 2000 (between 2000 and June 2025). For example, 97 enzymes (31.9% of all drug targets introduced since 2000) are modulated by 114 small molecules and 14 biologics. The data presented are just for small molecules and biologics,

so the total target counts for some families are lower than those in Fig. 1. Human and non-human targets are included. **b**, Distribution of potency values in small-molecule drugs against targets launched before and after the year 2000 (endogenous peptides are not shown because of the very small number of potency values). Significant differences ($p < 0.05$ according to the Brunner–Munzel test) between potencies of drugs approved pre and post 2000 for each target group are labelled. GPCR, G-protein-coupled receptor.

(Brunner–Munzel test; $p < 0.05$)²¹ more potent drugs in the post-2000 era compared with drugs approved before 2000.

This trend towards higher potency reflects advances in structure-based drug design, high-throughput screening technologies and a deeper understanding of target biology that have characterized modern medicinal chemistry.

Evolution during the past 25 years (2000–2024)

To assess the temporal dynamics of modern drug discovery, we examined annual drug approval trends over the past 25 years. Between

2000 and 2024, 1,091 drugs were approved by the FDA, EMA and PMDA, of which 826 (76%) have well-characterized MoAs with clearly defined targets. Annual drug approvals have increased, on average, over this period (Fig. 3a), rising from an average of 40 (s.d. = 8.6) in 2000–2004 to 59 (s.d. = 8.3) in 2020–2024. This increase was sustained by both the expansion of druggable targets and the growth of established drug classes with well-defined MoAs. There have been substantial fluctuations from year to year, with the first 2 years of the COVID-19 pandemic (2020 and 2021) being peak approval years, with 73 and 62 new drugs licensed, respectively, regardless of MoA elucidation^{22,23}.

In contrast, fewer than 30 drugs were approved in 2003, 2005 and 2010 (Fig. 3a).

Throughout this quarter century, the proportion of drugs with defined MoA targets has remained remarkably consistent at 76% (s.d. = 7.6%). This is consistent with trends reported in earlier

analyses¹, and indicates that the proportion of successful drug discovery programmes that begin with selection of a defined MoA target has remained broadly similar.

Analysis of newly introduced drug targets between 2000 and 2024 reveals 309 novel biomolecules, emerging at an average rate of

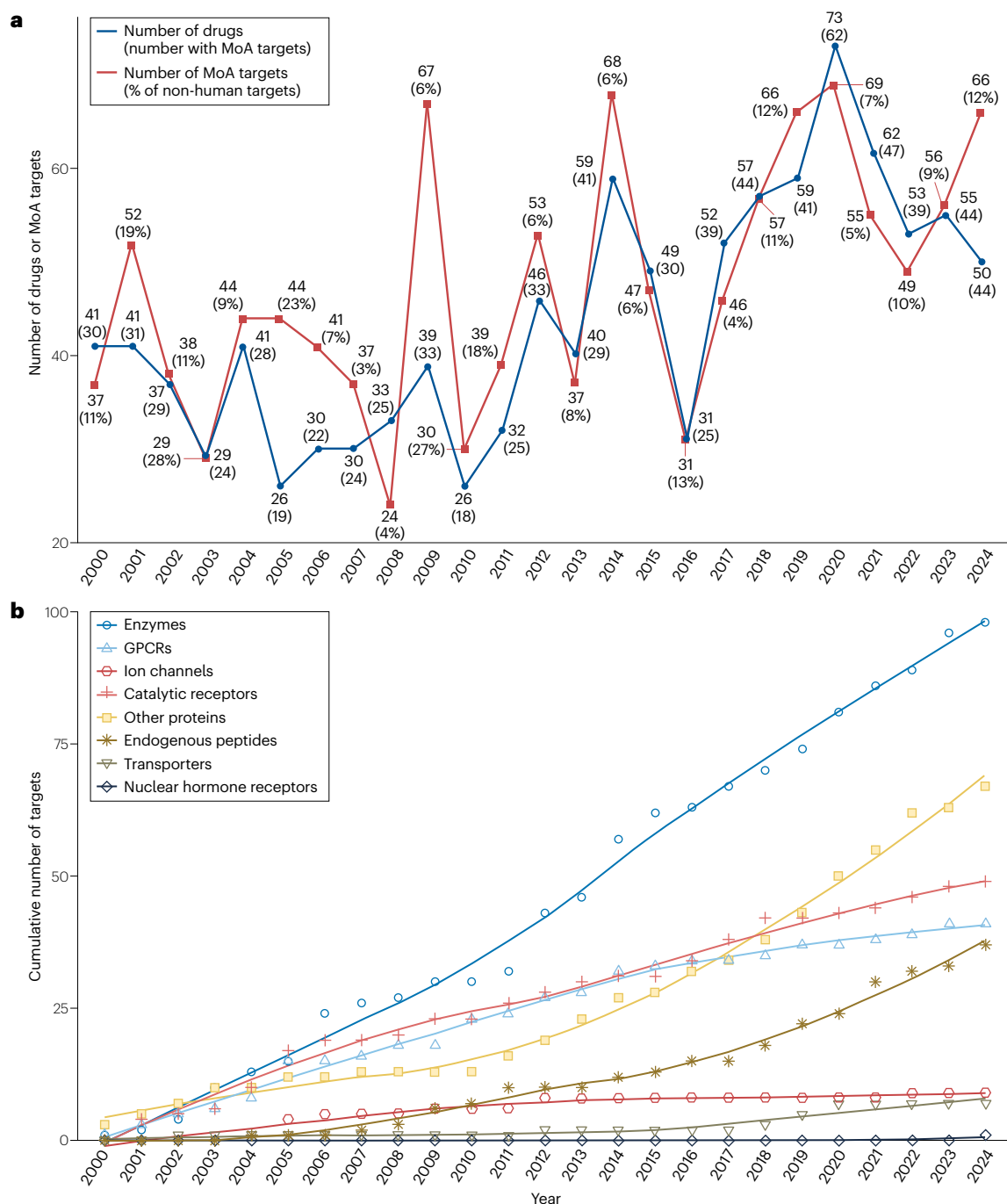


Fig. 3 | Trends for drug approvals and targets: 2000–2024. a, Annual counts of approved drugs (blue line), with the number of these drugs that have known mechanism of action (MoA) targets (in parentheses). The number of unique MoA targets for the drugs approved in each year (red line), with the percentage of

non-human MoA targets (from bacteria, viruses, fungi or parasites) (in parentheses). **b**, Cumulative annual increase in newly introduced MoA targets, categorized by protein family, such as enzymes, G-protein-coupled receptors (GPCRs) and ion channels.

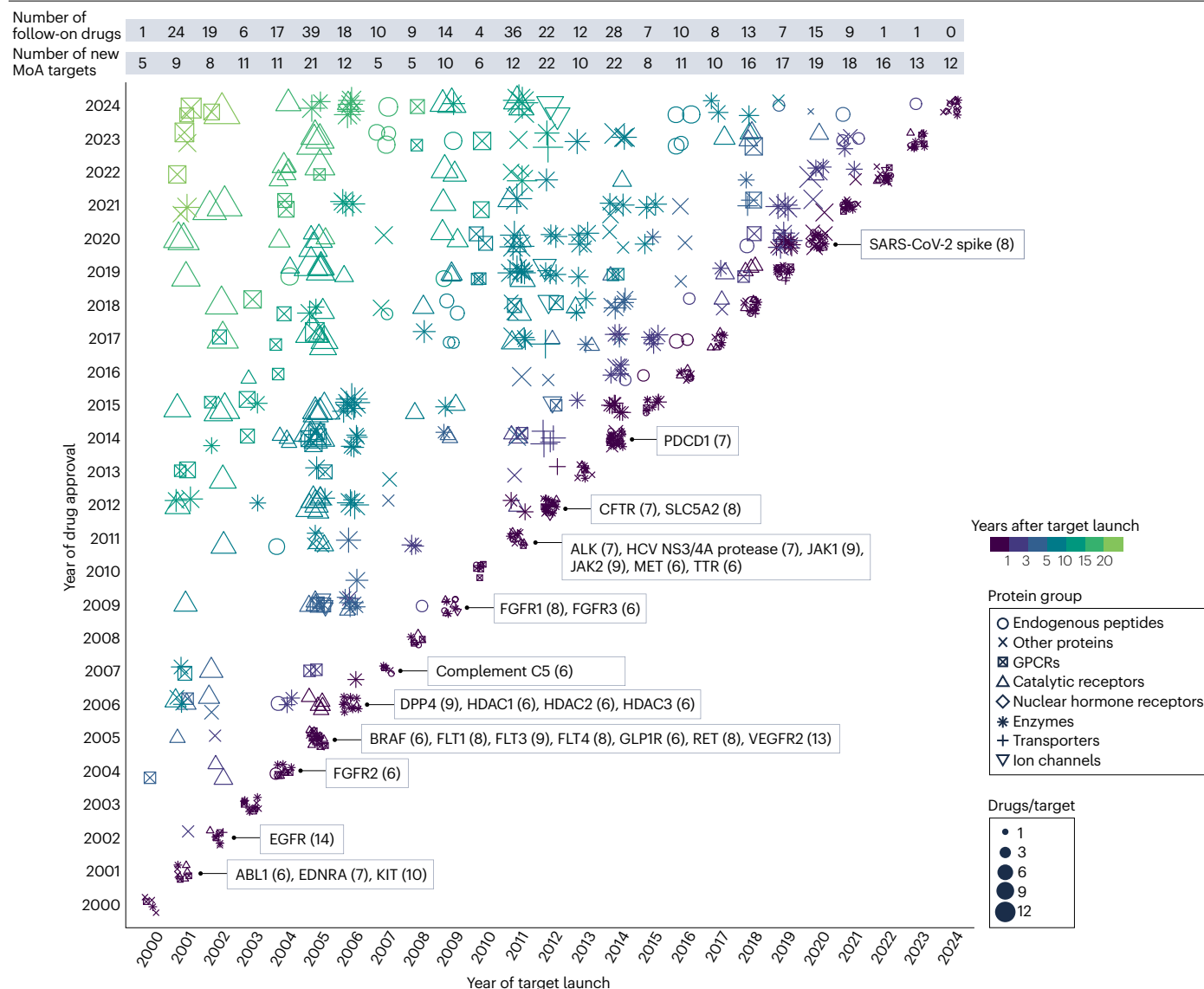


Fig. 4 | Trends in target launches with drug approvals: 2000–2024.

For each year, drug approvals with new mechanism of action (MoA) targets are shown on the diagonal of the graph using icons with shapes that indicate the protein group targeted. The number of these MoA targets is indicated at the top of the graph. Above the diagonal, the number (indicated by the size of the icon) and target type (indicated by the shape of the icon) for follow-on drugs for the targets are shown, with the total number of follow-on drugs for targets introduced in that year indicated at the top of the graph.

Targets with >5 follow-on drugs are highlighted below the diagonal, with the corresponding numbers of drugs (in parentheses). As an illustrative example, in 2002 epidermal growth factor receptor (EGFR), a catalytic receptor, was introduced as a drug target with the small-molecule kinase inhibitor gefitinib (approved in Japan), which has since (up to 2024) been followed by nine further small-molecule drugs and four antibodies, beginning with cetuximab in 2004. GPCR, G-protein-coupled receptor; VEGFR2, vascular endothelial growth factor receptor 2.

12 (s.d. = 5) targets per year. From a protein functional class perspective (Box 1), enzymes remain the dominant category among new targets, with steady and substantial year-on-year growth driven, in part, by novel kinase targets. Catalytic receptors, predominantly kinases, have also grown at a consistent rate.

Two groups of targets – endogenous peptides and those in the ‘other proteins’ category – showed sharp growth from about 2010 onwards (Fig. 3b), helping accelerate the overall rate of target innovation to 15 (s.d. = 5) novel targets annually through 2024. The surge in endogenous

peptide targets is driven by the approval of mAb therapeutics, as noted above. The recent expansion of unconventional drug targets (‘other proteins’) has been primarily fuelled by human cell-surface/immunological proteins.

In contrast, the pace of introducing novel GPCRs as drug targets has slowed down over the past decade, suggesting near-saturation of readily druggable family members. Ion channels, transporters and, particularly, nuclear hormone receptors also yielded very few newly validated drug targets during this period.

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Notably, transporters show signs of renewed interest in recent years (Fig. 3b).

The acceleration in the approval of drugs with novel MoA targets after 2010 can be attributed to several factors, including completion of the Human Genome Project, systematic exploration of the druggable kinome (aided by advances in structural biology and screening tools for chemical biology) and the maturation of technologies underlying biologic drugs. These advances are reflected not only in the emergence of new MoA targets but also in the progressive reuse of established MoAs, contributing to the growth of drug classes targeting the same proteins (Fig. 4). The most drugged targets are two catalytic receptors: EGFR (launched by gefitinib in 2002; 14 approved inhibitors, of

which 10 are small molecules and 4 are mAbs) and vascular endothelial growth factor receptor 2 (VEGFR2) (launched by sorafenib in 2005; 13 approved inhibitors, of which 12 are small molecules and 1 is a mAb), as illustrated in Fig. 4.

Important trends have also emerged in approved drug types. Whereas small molecules historically dominated pharmaceutical approvals, the twenty-first century has witnessed a dramatic rise in biologics. Comparing annual approvals, biologics (mAbs, ADCs, oligonucleotides, enzymes, fusion proteins and peptides) reached parity in 2023 (Fig. 5a). If current trends continue, biologics are likely to surpass small molecules in annual approval rates within the next few years. This expansion so far has been driven predominantly by mAbs and ADCs,

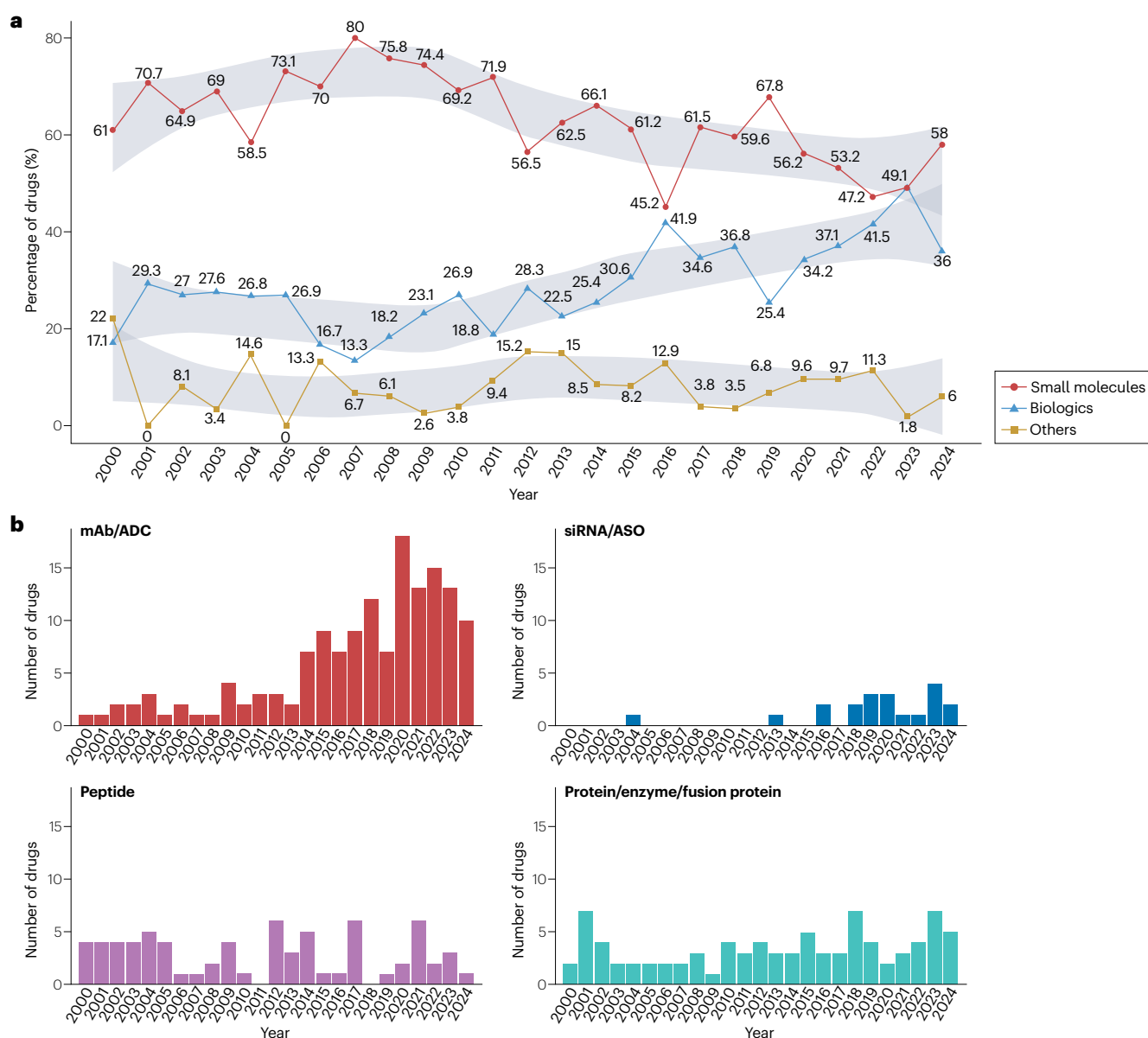


Fig. 5 | Trends in approved drug modalities and biologic subcategories: 2000–2024. a, Annual percentages of approved drugs classified as small molecules, biologics (including monoclonal antibodies (mAbs), oligonucleotides, proteins, enzymes, fusion proteins and peptides) and

other modalities (such as radiopharmaceuticals and inorganic compounds). **b**, Annual distribution of biologic drug approvals, broken down by subcategory. ADC, antibody–drug conjugate; ASO, antisense oligonucleotide; siRNA, small interfering RNA.

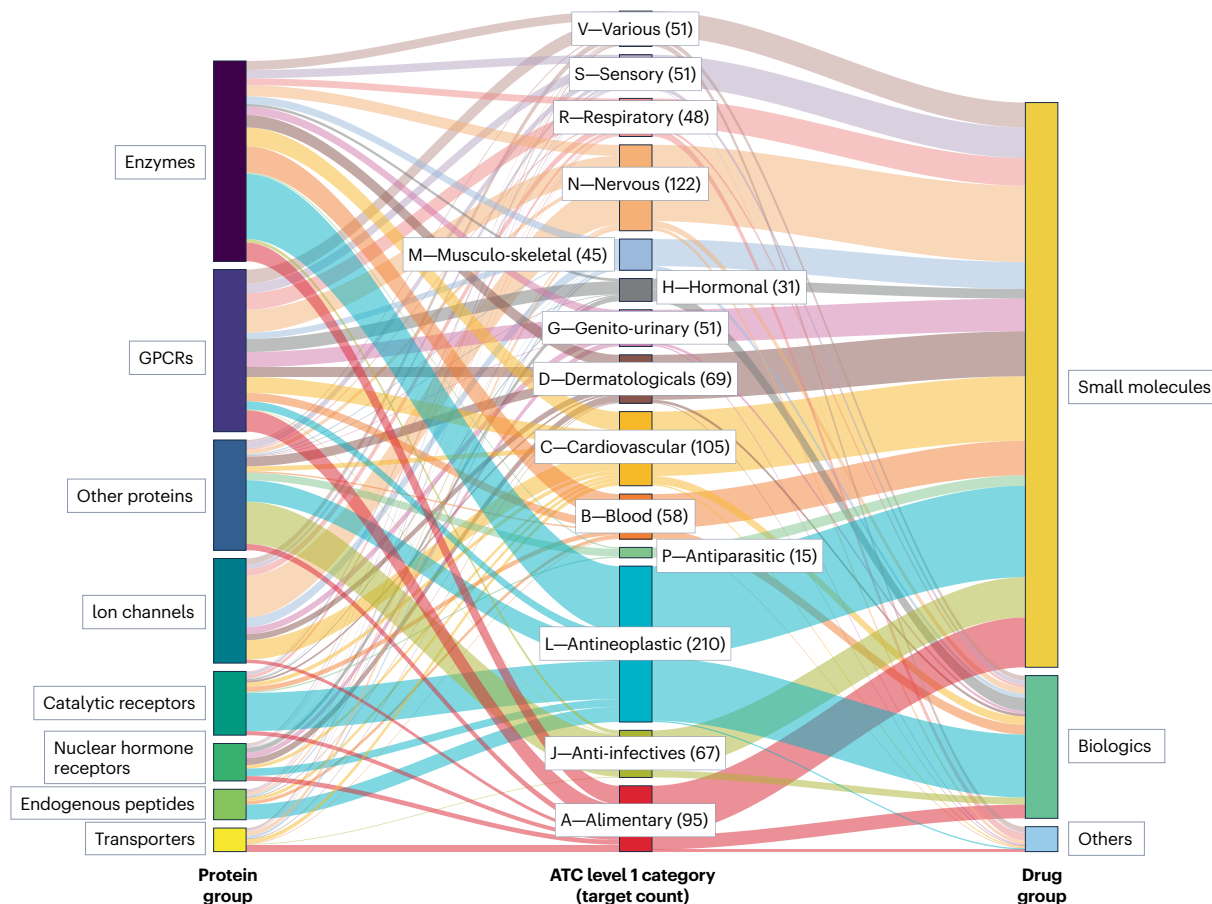


Fig. 6 | Distribution of protein target classes across therapeutic categories.

Sankey diagram showing the number of mechanism of action (MoA) targets in protein groups and the number of drugs in drug types linked to the 14 Anatomical Therapeutic Chemical (ATC) level 1 therapeutic categories, shown with abbreviated names. The full category names are: A, alimentary tract and metabolism; B, blood and blood forming organs; C, cardiovascular system; D, dermatologicals; G, genito-urinary system and sex hormones; H, systemic hormonal preparations, excluding sex hormones and insulins; J, anti-infectives for systemic use; L, antineoplastic and immunomodulating agents; M, musculo-skeletal system; N, nervous system; P, antiparasitic products, insecticides

and repellents; R, respiratory system; S, sensory organs; and V, various. Flow widths connecting protein groups to ATC categories are proportional to the number of protein targets within each therapeutic area. Flow widths connecting drug modalities to ATC categories reflect the number of approved drugs. Antineoplastic and immunomodulating agents (ATC class L) represent the largest therapeutic area, driven primarily by enzyme and catalytic receptor targets. This ATC category is the principal domain for biologics, and their number in this category is nearly equal to the number of small-molecule drugs in it. GPCR, G-protein-coupled receptor.

particularly since 2014, when they began accounting for 50–70% of all biologic approvals (Fig. 5b).

More recently, after decades of development challenges, oligonucleotide therapeutics such as antisense oligonucleotides (ASOs) and small interfering RNAs (siRNAs) have now established a firm position in the modern pharmacopoeia and represent a growing segment of the biologic landscape^{24,25} (Fig. 4b). Their MoAs, including sequence-specific mRNA degradation, splice modulation and post-transcriptional gene silencing, enable targeting of previously undruggable proteins and RNA transcripts, providing therapeutic options for genetic diseases, metabolic disorders and other conditions refractory to conventional approaches²⁴.

Currently, 22 oligonucleotide drugs have received regulatory approval, representing several mechanistically distinct classes. Seven siRNA therapeutics, which exploit the RNA interference pathway to

achieve sequence-specific mRNA degradation, have been approved: patisiran, givosiran, inclisiran, lumasiran, nedosiran, vutrisiran and fitusiran. Approved ASOs that function through RNase H-mediated degradation or steric blocking of a specific mRNA are another important category, which includes fomivirsin, mipomersen, volanesorsen, tofersen, inotersen and eplontersen. Other approved ASOs modulate pre-mRNA splicing to restore functional protein production, and include nusinersen, eteplirsin, golodirsin, viltolarsen and casimersen. The two approved aptamers, pegaptanib and avacincaptad pegol, fold into complex 3D structures and block protein function in a manner analogous to mAbs. Finally, the Toll-like receptor 9 agonist HepB-CpG is used as an immunostimulatory adjuvant in hepatitis B vaccination, and defibrotide, a mixture of porcine-derived single-stranded oligonucleotides with complex antithrombotic and profibrinolytic properties, is approved for hepatic veno-occlusive disease²⁶.

Therapeutic areas

To understand the therapeutic impact of the innovations described above, we examined the distribution of drug targets across therapeutic areas and how different protein classes contribute to specific therapeutic applications.

We mapped 1,526 drugs (90% of our dataset) to 14 therapeutic areas defined by the level 1 categories of the World Health Organization (WHO) Anatomical Therapeutic Chemical (ATC) classification system (Box 1). These drugs modulate 656 MoA protein targets, representing 96% of all efficacy targets identified in this study.

The distribution of protein target classes across ATC therapeutic categories is illustrated in Fig. 6 using a Sankey diagram. Because individual drugs may be assigned multiple ATC codes spanning different therapeutic indications, the visualization captures this therapeutic versatility through the overlapping distribution of protein targets across categories. This multiplicity reflects the reality that many drugs exhibit efficacy in multiple disease contexts, either through intentional polypharmacology or through repurposing for additional indications after initial approval.

Antineoplastic and immunomodulating agents (ATC code L) represent the therapeutic area with the largest number of both drug targets

Box 2 | Temporal dynamics between compound registration and market authorization

Compound disclosure, approximated by CAS Registry Number assignment, was used to estimate development timelines from initial disclosure to regulatory approval, following the methodology described in Box 1. Disclosure practices may vary among drug modalities, affecting the accuracy of the development timelines.

Among 613 small molecules, 90% were registered after 1983, and 80% were approved within 18 years of registration. For biologics, 90% of the 325 drugs analysed were registered after 1995, and 80% were approved within 10 years (see the figure, panel a). Median approval times reveal substantially shorter timelines for biologics (6 years) compared with small molecules (11 years) (see the figure, panel b).

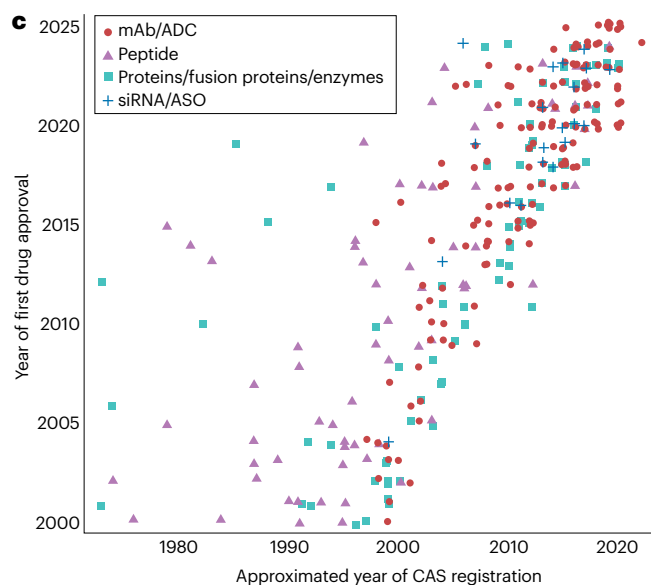
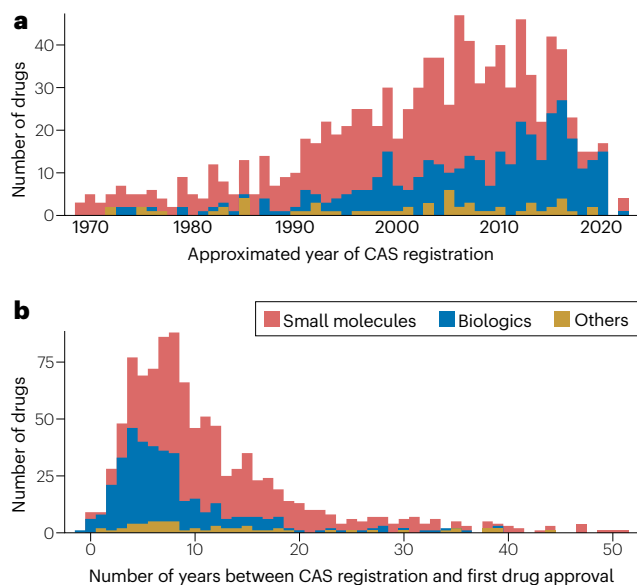
Within biologics, peptide drugs have notably longer development timelines, averaging about 12 years from CAS registration to approval (comparable with small molecules). In contrast, other biologic modalities, including monoclonal antibodies (mAbs), RNA-based therapeutics and replacement proteins, typically achieve approval within 5–6 years (see the figure, panel c), probably due to later-stage disclosure of advanced biologic candidates.

Development timelines vary widely. Some drugs are registered and approved in the same year, including the small molecule eluxacaptor

(2019), the mAbs imdevimab, casirivimab, bamlanivimab and etesevimab (all 2020), and the peptide peginesatide (2012). Conversely, extended timelines occur when compounds are shelved and later repurposed: the small molecules clascoterone and nifurtimox were registered circa 1970 but approved approximately 50 years later, whereas the peptide afamelanotide (registered in about 1981) was approved in 2014.

For first-in-class drugs acting on novel targets, development timelines vary by modality. Of 330 drugs targeting new proteins, 80% received CAS registration within 15 years of approval: 15 years for small molecules versus 11 years for biologics. The average timeline for these pioneering therapeutics is approximately 10 years (see Supplementary information).

These findings reveal distinct temporal patterns that reflect differences in disclosure practices, development complexity and regulatory pathways across drug modalities. Notably, biologics (particularly advanced modalities such as mAbs and RNA therapeutics) demonstrate accelerated development timelines. At the same time, first-in-class drugs targeting novel mechanisms maintain relatively consistent timelines regardless of modality. ADC, antibody–drug conjugate; ASO, antisense oligonucleotide; siRNA, small interfering RNA.



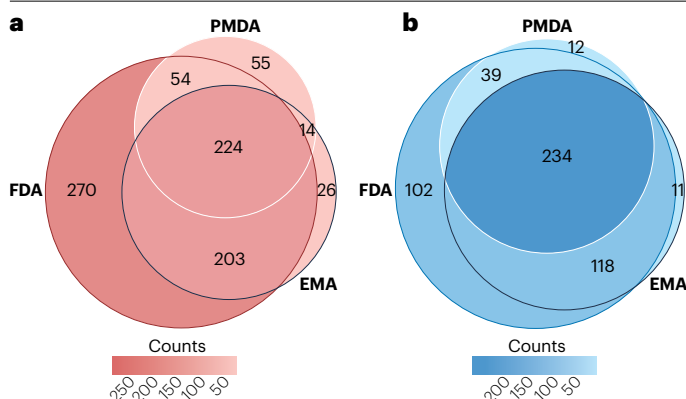


Fig. 7 | Overlap of drugs and their targets for new drugs approved by three major regulatory agencies between 2000 and June 2025. a, Overlap between new drugs approved by the US Food and Drug Administration (FDA), European Medicines Agency (EMA) and Japanese Pharmaceutical and Medical Devices Agency (PMDA). **b**, Overlap between the mechanism of action (MoA) targets for the new drugs approved by the three regulatory agencies.

($n = 210$) and approved drugs ($n = 322$). This category is dominated by enzymes, catalytic receptors and ‘other proteins’ (Fig. 6). Notably, 70% of all catalytic receptor targets and half of all endogenous peptide targets are utilized within this therapeutic area, reflecting the central role of growth factor and cytokine signalling in cancer and immune disorders²⁷.

Biologics have achieved exceptional penetration in oncology and immunology, accounting for 42% of drugs approved for ATC code L, nearly matching the 57% represented by small molecules. Overall, 45% of all approved biologics are indicated for haematologic, oncologic or immunologic conditions. This high proportion reflects the suitability of antibody-based therapeutics for targeting extracellular proteins such as immune checkpoints, tumour antigens and cytokines that are much less tractable to target with small molecules.

Biologics also dominate systemic hormonal preparations (ATC code H), where they account for 54% of drugs and modulate 61% of targets within this therapeutic area (Fig. 6). This is not surprising given the inherent compatibility between peptide hormones and biologic drug platforms, enabling direct replacement or antagonism of endogenous hormones.

Nervous system disorders (ATC code N) constitute the second-largest therapeutic area by target count, with 122 proteins modulated by 260 drugs, of which 94% are small molecules (Fig. 6). This category is predominantly composed of ion channels and GPCRs, which mediate neurotransmitter signalling and membrane excitability in neurological and psychiatric disorders.

Cardiovascular diseases (ATC code C) rank third, with 105 protein targets and 187 approved drugs. Drugs in this therapeutic area modulate a more diverse array of target classes, with GPCRs, enzymes and ion channels representing the most prevalent groups (Fig. 6). The breadth of target classes illustrates the multifactorial nature of cardiovascular pathophysiology, encompassing haemodynamic regulation, thrombosis, lipid metabolism and cardiac electrophysiology. Additional details on target distribution across all therapeutic areas are provided in Supplementary information.

Kinase inhibitors have demonstrated exceptional utility in oncology, based on the key role of kinases in growth factor signalling

pathways that promote malignant transformation and tumour progression. Of the 120 kinase inhibitors assigned ATC codes, 94 drugs (78%), including 72 small molecules, are approved for cancer treatment (ATC code L01). These agents target 49 kinases (73% of all drugged kinases), comprising 25 catalytic receptors and 24 enzymes. A comprehensive list of approved kinase inhibitors and their targets is provided in Supplementary Table 4.

Innovation in global markets

Although therapeutic area analysis reveals how targets and drugs distribute across diseases, understanding the geography and timelines of innovation provides additional insights into the forces shaping drug discovery. Regulatory frameworks, development timelines and geographic concentration of innovation all influence which targets get validated and how quickly new therapeutics reach patients.

The trajectory from newly discovered compounds to regulatory approval is a complex, highly regulated process spanning multiple stages. In the typical target-based drug discovery approach, following hit identification and lead optimization against a defined protein target, promising candidates undergo preclinical testing in animal models to assess safety and efficacy. Successful candidates then advance through sequential clinical trials (phase I, phase II and phase III), evaluating safety, dosing and therapeutic benefit in progressively larger patient populations. Upon positive outcomes, sponsors submit comprehensive data packages to regulatory agencies, which conduct rigorous reviews before granting marketing approval.

The timeline from initial disclosure to approval varies substantially by drug modality (Box 2). For small-molecule drugs approved after 2000, the median development time is approximately 11 years from CAS registry number assignment, with 80% achieving approval within 18 years. In contrast, biologics demonstrate notably shorter development timelines, with 80% approved within 10 years of initial disclosure. In general, biologics show higher target specificity and more predictable pharmacology²⁸, and the regulatory frameworks have evolved to accommodate these therapeutics, facilitating expedited timelines.

For drugs introducing novel efficacy targets (first-in-class therapeutics approved during the past 25 years), the average time from first public disclosure to regulatory approval is 10 years, with 80% approved within 15 years (Box 2). In many cases, such drugs carry an additional validation burden associated with pioneering new therapeutic mechanisms, where both target biology and clinical benefit must be established *de novo*.

Drug approval dynamics differ substantially among the three major regulatory agencies examined: the FDA, EMA and PMDA. Since 2000, the FDA has approved 750 drugs with well-defined MoA proteins, compared with 466 by the EMA and 347 by the PMDA. Surprisingly, only 27% of these drugs have received approval from all three agencies

Table 1 | Number of drugs by first market approval between 2000 and June 2025

Agency	Small molecules	Biologics	Others	Total
FDA	402	200	15	617
EMA	67	53	3	123
PMDA	81	23	2	106

EMA, European Medicines Agency; FDA, US Food and Drug Administration; PMDA, Pharmaceuticals and Medical Devices Agency.

(Fig. 7). Collectively, these represent 845 unique drugs, with more than 80% ($n = 616$) receiving first approval in the United States (Table 1).

Of the drugs first approved by the FDA, 44% ($n = 269$) had not received marketing authorization from either the EMA or the PMDA within the time frame analysed. Among the remaining 348 FDA-first drugs that eventually gained approval elsewhere, 67% were approved by the EMA and 38% by the PMDA within 1 year of their US launch. The leading position of the United States in new drug approvals is particularly evident for kinase inhibitors, a hallmark of modern targeted therapy: 83% ($n = 105$) received their initial regulatory approval from the FDA. Novel drug approvals outside the United States, Japan and the European Union – particularly in China – are attracting increasing attention (Box 3).

The prominence of the US market extends to target innovation. Of the 312 novel MoA drug targets introduced since 2000, 78% were first validated through FDA-approved drugs, with drugs approved first by the EMA and the PMDA each contributing 11% of novel targets. On an annual basis, FDA approvals account for 60–100% of newly introduced MoA targets (see Supplementary information). Across most protein families, FDA approvals drove target expansion, introducing approximately 70% of novel MoA targets within each class. Transporters represent a notable exception: of seven novel transporter targets, the FDA introduced three (43%), whereas the EMA and PMDA contributed three

and two, respectively (see Supplementary information). Finally, pathogenic (non-human) drug targets were introduced almost exclusively through FDA approvals. Of 26 such targets launched during this period, 21 (81%) were first validated by FDA-approved drugs.

Overall, these findings underscore the dominant role of the US market in shaping the contemporary therapeutic target landscape, supported by robust regulatory infrastructure, substantial investment in biomedical research and innovation-incentivizing policies that accelerate both drug development and regulatory review. In addition, strategic market prioritization and regulatory timing might play a role.

Conclusion and outlook

This comprehensive analysis of drug targets from 2000 to 2025 reveals a fundamentally transformed landscape of protein druggability. The expansion in novel targets demonstrates that the druggable proteome extends well beyond the classical receptor, enzyme and ion channel paradigm that dominated twentieth-century pharmacology. This expansion has been driven by several forces: genomic and structural biology advances that support rational target identification, advances in small-molecule drug discovery enabling selectivity and potency for targets in key families such as kinases, the maturation of antibody-based therapeutic platforms, and the emergence of RNA-based modalities that can access previously intractable targets.

Box 3 | Drug target innovation in China: 2019–2023

A recent analysis by Yan et al. compared new drug approvals by regulatory agencies in China³⁹ (National Medical Products Administration (NMPA)), Japan, the United States and Europe between 2019 and 2023. The NMPA approved 256 new drugs, representing 5.4% more approvals than the US Food and Drug Administration (FDA), 34% more than the European Medicines Agency (EMA) and 37% more than the Pharmaceuticals and Medical Devices Agency (PMDA).

Using the list of drugs with Chinese approval dates provided in the supplementary material of that study³⁹, we evaluated target innovation. We first retained NMPA approvals from 2019 to 2023 and split combination therapies into individual agents, yielding 216 drugs approved in China over these 5 years. Of these, 63 had not been approved by the FDA, EMA or PMDA by June 2025 and collectively introduced six novel targets (see the table).

Of the drugs also approved by other agencies, 24 lacked well-established efficacy targets, leaving 127 drugs with defined

mechanisms of action approved in China and at least one additional jurisdiction. Among these, efbemalenograstim alfa, sugemalimab, serplulimab and ensartinib were first launched in China; however, none were first-in-class for their targets. Overall, during these 5 years, target innovation in China lagged behind other regions, with 59 new drug targets being introduced in the United States, 12 in Japan and 11 in Europe.

These observations are consistent with an independent analysis of oncology drug approvals⁴⁰ between 2015 and 2021, which reported that 32% of 90 FDA-approved oncology drugs were first-in-class, compared with none in China. Conversely, many drugs initially approved in the United States were later approved in China, with the approval lag decreasing to less than 4 years after FDA approval⁴⁰. During the revision of this Review, the latest article in the Target Watch series was published, reviewing novel drug targets in 2025 (ref. 41). For the first time, NMPA approvals were added to the analysis, including the first drugs targeting cathepsin L and tetanus toxin.

Pioneering drug targets of products approved by the Chinese NMPA: 2019–2023

Target name	Gene	Drug name (year)	Indication
Tumour necrosis factor receptor superfamily member 10A ^a	TNFRSF10A	Aponermin (2023)	Multiple melanoma
Tumour necrosis factor receptor superfamily member 10B ^b	TNFRSF10B	Aponermin (2023)	Multiple melanoma
Tumour necrosis factor ligand superfamily member 13 ^c	TNFSF13	Telitacicept ^d (2021)	Systemic lupus erythematosus
Peroxisome proliferator-activated receptor- δ	PPAR- δ	Chiglitazar (2021)	Type 2 diabetes
Hexokinase-4 ^e	GCK	Dorzagliatin (2022)	Type 2 diabetes
RABV-G protein	RABV-G protein	Ormutivimab (2022)	Rabies

^aAlso known as death receptor 4. ^bAlso known as death receptor 5. ^cAlso known as a proliferation-inducing ligand (APRIL). ^dAlso inhibits BAFF/BlyS. ^eAlso known as glucokinase.

The rise of biologics represents the most consequential shift in modern drug discovery. The post-2000 target distribution shows how antibody-based therapeutics have enabled extracellular signalling proteins to be successfully targeted, particularly endogenous peptides and catalytic receptors. However, biologics remain constrained by cellular access barriers. This complementarity between

modalities has been crucial: whereas protein-based biologics excel at targeting cell-surface and secreted proteins, small molecules retain unique advantages for intracellular and membrane-embedded targets, including intracellular enzymes, nuclear hormone receptors, ion channels and transporters. RNA therapeutics have emerged as a third pillar, enabling sequence-specific modulation of some targets that have been

Box 4 | Cell and gene therapies

As the presented analysis excludes cell and gene therapies (CGTs), we provide a list of CGT-associated targets (see the table) that are not targeted by other modalities. Many gene/cell therapies act by gene addition/replacement, where the ‘target’ is a deficient gene/protein being supplied rather than antagonized; others act by genome editing or by redirecting immune cells against a surface antigen as described below.

The 14 targets listed in the table span several protein classes. Enzymes represent the largest group (ACPP, ARSA, F8, RPE65), followed by endogenous peptides used as locally delivered cytokines (CNTF, CSF2, IFNA2), and transporters (ABCD1, OTOF). The remaining targets include a transcription factor (BCL11A), a structural extracellular matrix protein (COL7A1), haemoglobin subunit-β (HBB), a cancer-testis antigen (MAGEA4) and a cytoskeletal signalling scaffold (WAS).

Across these 14 targets, three broad mechanistic rationales explain why CGT modalities were required. First, the majority involve gene replacement, where a functional copy of a deficient gene is delivered to correct a monogenic disorder. This applies to ARSA and ABCD1 (lysosomal and peroxisomal enzymes whose replacement requires intracellular trafficking to specific organelles, beyond the reach of systemic enzyme replacement therapy), F8 (coagulation factor requiring sustained hepatic expression), RPE65 (a retinal pigment epithelium enzyme sequestered behind the blood–retina

barrier), HBB (a structural protein requiring erythroid-lineage expression), WAS (a haematopoietic-specific cytoskeletal regulator), OTOF (an inner-ear synaptic protein required for auditory signal transmission) and COL7A1 (an anchoring fibril collagen at the dermal–epidermal junction). Second, BCL11A represents a distinct paradigm: rather than supplying a missing gene, CRISPR–Cas9 gene editing disrupts a transcriptional repressor of fetal haemoglobin to treat sickle cell disease. Last, some targets serve as antigens for immune-directed therapies (MAGEA4 for TCR-engineered T cells, ACPP for dendritic cell immunotherapy) or as locally expressed cytokines delivered via viral vectors (CSF2 via oncolytic HSV-1, IFNA2 via an adenoviral vector, CNTF via encapsulated cell technology).

A common thread is that these targets are inaccessible to conventional drug modalities. Many are intracellular proteins (ARSA, ABCD1, WAS, HBB, RPE65, BCL11A, OTOF) that cannot be engaged by monoclonal antibodies; several reside in anatomically privileged compartments (RPE65 behind the blood–retina barrier, OTOF in the cochlea, ARSA and ABCD1 in the central nervous system), where systemic biologics do not achieve therapeutic concentrations; and others require sustained or lineage-restricted expression (F8, HBB) that transient small molecules or oligonucleotides cannot provide. These constraints collectively define the therapeutic space uniquely addressed by CGT modalities.

Exclusive drug targets for cell and gene therapies

Group name	Target name (gene)	Drug name (commercial name)
Endogenous peptides	Ciliary neurotrophic factor (CNTF)	Revakinagene taroretcel (Encelto)
Endogenous peptides	Granulocyte–macrophage colony-stimulating factor (CSF2)	Talimogene laherparepvec (Imlygic)
Endogenous peptides	Interferon-α ₂ (IFNA2)	Nadofaragene firadenovec (Adstiladrin)
Enzymes	Prostatic acid phosphatase (ACPP)	Sipuleucel-T (Provenge)
Enzymes	Arylsulfatase A (ARSA)	Atidarsagene autotemcel (Lenmeldy)
Enzymes	Retinoid isomerohydrolase (RPE65)	Voretigene neparvovec (Luxturna)
Enzymes	Coagulation factor VIII (F8)	Valoctocogene roxaparvovec (Roctavian)
Other proteins	BCL11 transcription factor A (BCL11A)	Exagamlogene autotemcel (Casgevy)
Other proteins	Collagen-α1(VII) chain (COL7A1)	Beremagene geperpavec (Vyjuvek), Prademagene zamikeracel (Zevaskyn)
Other proteins	Haemoglobin subunit-β (HBB)	Betibeglogene autotemcel (Zynteglo), Lovotibeglogene autotemcel (Lyfgenia)
Other proteins	Melanoma-associated antigen 4 (MAGEA4)	Afamitresgene autoleucel (Tecelra)
Other proteins	Wiskott–Aldrich syndrome protein (WAS)	Etuvetidigene autotemcel (Waskyra)
Transporters	Otoferlin (OTOF)	Lunsotogene emelioparvovec (Otarmeni)
Transporters	ATP-binding cassette sub-family D member 1 (ABCD1)	Elivaldogene autotemcel (Skysona)

Glossary

Antibody–drug conjugates

(ADCs). Bioconjugates comprising a monoclonal antibody (mAb) linked to a cytotoxic payload, enabling targeted delivery of the payload to antigen-expressing cells.

Antisense oligonucleotides

(ASOs). Short, single-stranded nucleic acids that bind complementary mRNA sequences to block translation, promote degradation or promote exon skipping.

Biologics

For the analysis in this Review, biologics include antibodies, oligonucleotides, other proteins, enzymes, fusion proteins and peptides, and exclude vaccines and gene or cell therapies.

Drug

A therapeutic agent used to treat, prevent, diagnose or alleviate symptoms of disease.

Drug potency

The concentration of a drug required to elicit a defined biological effect, commonly reported as IC_{50} , EC_{50} , K_d or K_i . Here, potency is expressed as $-\log_{10}[M]$, where higher values denote greater potency (for example, a value of 9 corresponds to 1 nM).

First-in-class drugs

In the context of this Review, defined as drugs that have mechanism of action targets that have not been previously modulated by approved drugs.

Fusion proteins

Chimeric proteins engineered to combine functional domains or targeting elements from different proteins.

Mechanism of action (MoA) targets

(Also known as efficacy targets). In this Review, defined as proteins (such as an enzyme, G-protein-coupled receptor (GPCR), transporter or ion channel) or other biomolecules (such as mRNA, pre-mRNA, DNA or peptide) directly bound by a drug to elicit its therapeutic effect. Targets related to absorption, distribution, metabolism, excretion and toxicity (ADMET) or those lacking validated clinical relevance were excluded. A full list of mechanism of action (MoA) targets is provided in Supplementary Tables 1 and 2.

Monoclonal antibodies

(mAbs). Antibodies derived from a single clone of immune cells, designed to bind specific antigens on target cells.

Peptides

Therapeutics composed of short chains of amino acids (typically <40 residues).

Radioligands

Radiolabelled molecules used for therapeutic or diagnostic purposes, enabling targeted delivery of radionuclides or imaging of disease-relevant cells.

Small interfering RNAs

(siRNAs). Synthetic short RNAs designed to specifically target and degrade mRNA transcripts, thereby inhibiting the expression of specific genes.

Small-molecule drugs

Organic therapeutic agents with molecular mass below 1,250 Da. Exceptions include fondaparinux and pafolacianine, with molecular masses above 1,250 Da.

intractable for both small molecules and protein-based biologics. The faster development timelines for biologics (median 6 years from disclosure) compared with small molecules (median 11 years) has been supported by their higher target specificity and more predictable pharmacology, as well as the adaptation of regulatory frameworks to accommodate these novel therapeutic modalities.

The geographic concentration of the regulatory approvals for innovative medicines remains striking, with the majority of first-in-class drug approvals that introduce novel MoA targets occurring in the United States. This leading position reflects synergistic effects between regulatory infrastructure, research investments and commercial incentives that accelerate both development and validation of new therapeutic mechanisms.

The sustained increase in annual drug approvals and target diversity suggests continued expansion. However, plateaux in certain target classes indicate that although the druggable proteome is expanding, fundamental limits may be approaching. Whether these represent biological constraints or merely technological boundaries will define the next era of pharmaceutical innovation.

Several key trends might shape the next phase in the evolution of the target landscape. Emerging modalities, including targeted protein degraders such as proteolysis-targeted chimeras (PROTACs), molecular glues, macrocyclic drugs, peptide–drug conjugates and CRISPR-based therapeutics (Box 4), promise to access targets that have historically been considered undruggable. Artificial intelligence-driven discovery could accelerate target identification and validation²⁹. Integration of multi-omic datasets may shorten the 10-year average timeline for first-in-class drugs and improve clinical success rates.

Overall, the trends discussed in this Review highlight a dynamic drug discovery ecosystem that is transitioning towards more precise,

potent and diverse modalities. This ecosystem has also demonstrated high adaptivity in recent years, with the pace of development of new modalities to address previously undrugged targets and established targets – including those that have historically been considered ‘undruggable’ – accelerating. Future research and development efforts should continue to leverage this expanding target space while addressing the challenges of drug delivery, safety and patient stratification, serving as a gateway to personalized medicine. The accrual, sharing and understanding derived from on-target and off-target structure–activity relationships and toxicology knowledge will increasingly inform the design of newer and safer therapeutic agents, particularly when combined with validated artificial intelligence technologies. Ultimately, this evolving landscape has the potential to deliver transformative therapies for complex and unmet medical needs.

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Author contributions

T.I.O., L.H., S.A., R.C., J.P.O., B.A.-L., S.S., A.P. and N.S. researched data for the article. All authors contributed substantially to discussion of the content. All authors wrote the article. T.I.O., L.H., S.A. and A.R.B. reviewed and/or edited the manuscript before submission.

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Competing interests

J.P.O. is an employee of Xyme AI. S.S. is an employee of Expert Systems. A.P. and N.S. hold financial interests in Torrey Pines Investments. A.R.B. and T.I.O. are employees of Dompé farmaceutici. B.A.-L. holds financial interests in Xyme, AstraZeneca and Recursion Pharmaceuticals following its acquisition of Exscientia; has served on scientific advisory boards for and/or received consulting fees from Astex Pharmaceuticals, AstraZeneca, GSK, Novo Nordisk, Santé Ventures and Astellas Pharma; and also holds or has held board, leadership or scientific advisory roles with Blood Cancer United, Cancer Research UK, the Chemical Probes Portal, the New York Genome Consortium and Open Targets. T.I.O. was an employee of Roivant Sciences and Expert Systems, and had financial interests in Roivant Sciences, Expert Systems and InSilico Medicine when this Review was written. The other authors declare no competing interests.

Additional information

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